THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $x = 1$ is a root of $kx^2 + 3x - 5 = 0$, then $k =$ 1

(a) -8 **(b)** 2 **(c)** 5 **(d)** -2

2. Which of these is a rational number? 1

(a) $2\sqrt{5}$ **(b)** $\sqrt{4}$ **(c)** $\dfrac{\sqrt{12}}{\sqrt{3}}$ **(d)** $\sqrt{2} + \sqrt{3}$

3. For the cubic with zeroes $1, 2, 3$, $\alpha\beta + \beta\gamma + \gamma\alpha$ is: 1

(a) 11 **(b)** 6 **(c)** -11 **(d)** 36

4. If $x + y = 7$ and $xy = 10$, then $x^2 + y^2 =$ 1

(a) 39 **(b)** 49 **(c)** 29 **(d)** 69

5. The equation $kx^2 + 6x + 3 = 0$ has real roots when: 1

(a) $k \le 3$ and $k \neq 0$ **(b)** $k < 3$ only

(c) $k \ge 3$ **(d)** $k > 0$

6. A number ends with exactly 2 zeros if its prime factorisation (other factor 3) is: 1

(a) $2^3 \times 5^1 \times 3$ **(b)** $2^3 \times 5^2 \times 3$ **(c)** $3^2 \times 7$ **(d)** $2^1 \times 5^2 \times 3$

7. If the zeroes of $x^2 - 10x + k$ differ by 4, then k is: 1

(a) 29 **(b)** 21 **(c)** 84 **(d)** 42

8. If $5x + 3y = 47$ and $3x + 5y = 41$, then $x - y =$ 1

(a) $\frac{3}{4}$ **(b)** 3 **(c)** 6 **(d)** 44

9. If α, β are roots of $x^2 - 5x + 6 = 0$, then $\dfrac{1}{\alpha} + \dfrac{1}{\beta} =$ 1

(a) 30 **(b)** $\frac{5}{6}$ **(c)** $-\frac{5}{6}$ **(d)** $\frac{6}{5}$

10. The LCM of 36, 48 and 72 is: 1

(a) 1728 **(b)** 72 **(c)** 5184 **(d)** 144

11. If $x = 2$ is a zero of $x^2 - 9x + 14$, the other zero is: 1

(a) 12 **(b)** -7 **(c)** 11 **(d)** 7

12. The pair $kx + 3y = k - 3$ and $12x + ky = k$ has no solution when $k =$ 1

(a) -6 **(b)** 12 **(c)** 3 **(d)** 6

13. **Assertion (A):** The equation $2x^2 - 3x + 5 = 0$ has no real roots. 1

Reason (R): If the discriminant $b^2 - 4ac < 0$, a quadratic equation has no real roots.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** The number $5 - 2\sqrt{3}$ is irrational. 1

Reason (R): The sum of a rational number and an irrational number is irrational.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** The sum of the zeros of $2x^2 - 8x + 6$ is 4. 1

Reason (R): For $ax^2 + bx + c$ with zeros α, β, the sum of the zeros is $\frac{c}{a}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. If the sum of the zeroes of $x^2 - 6x + k$ is equal to their product, find k. 2

17. Solve: $5x + 3y = 47$ and $3x + 5y = 41$. 2

18. If α, β are the roots of $x^2 - 6x + 5 = 0$, find $\alpha^2 + \beta^2$. 2

19. Prove that $3\sqrt{7}$ is irrational, given that $\sqrt{7}$ is irrational. 2

20. If one zero of $x^2 - 8x + k$ is 3 times the other, find k. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. For what values of k does $x^2 + 5x + k = 0$ have real roots? Find the greatest integer value of k. 3

22. Is $5 \times 6 \times 7 + 7$ a composite number? Justify. 3

23. If α, β, γ are the zeroes of $x^3 - 6x^2 + 11x - 6$, find $\alpha^2 + \beta^2 + \gamma^2$. 3

24. Find the area of the triangle formed by the lines $x + y = 6$, $x - y = 2$ and the x-axis. 3

OR

The sum of the squares of two consecutive odd positive integers is 74. Find them. 3

25. A, B and C take 20, 28 and 35 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

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OR

Find a quadratic polynomial whose zeroes are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, where α, β are zeroes of $x^2 - 5x + 6$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Meera is designing a parabola-shaped flower bed in her school garden. She models its boundary by the graph of $y = p(x) = x^2 - 8x + 15$, where x is measured in metres along the ground. 5

(i) Find the zeros of $p(x)$.

(ii) Find the sum and the product of the zeros.

(iii) How many times does the graph of $y = p(x)$ meet the x-axis, and at which points?

(iv) Form a quadratic polynomial whose zeros are the reciprocals of the zeros of $p(x)$.

27. A man travels 360 km partly by train and partly by bus. If he travels 150 km by train and the rest by bus, he takes $\frac{41}{4}$ hours; if he travels 240 km by train and the rest by bus, he takes 11 hours. Find the speeds of the train and the bus. 5

OR

A motorboat goes 48 km upstream and 48 km downstream in 7 hours. If the speed of the stream is 2 km/h, find the speed of the boat in still water. 5

END OF QUESTION PAPER **Total: 50 Marks**

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